Prajwal Halkude

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.50
Diamond PU science college Bhalki (585 328)	Class XII (Department of Pre-University Education karnataka)	2021	83.50%
Diamond High school, Bhalki (585 328)	Class X (Karnataka Secondary Education Examination Board)	2019	91.20%

SCHOLASTIC ACHIEVEMENTS

- **University Gold Medalist - Chemical Engineering, Visvesvaraya Technological University (VTU)** -- Awarded the **University Gold Medal** for outstanding academic performance, presented by the **Hon'ble Governor of Karnataka**.
- Winner, **Essay** and **Speech** Competition at a BIS (Bureau of Indian Standards) event.

PROJECTS

Power BI Ride Booking Analytics Dashboard:

- Built an interactive Power BI ride-booking analytics dashboard covering booking trends, revenue performance, vehicle utilization, cancellations, customer/driver ratings, and service-quality metrics across multiple analytical views.
- Developed business intelligence KPIs including customer retention rate, cancellation percentage, revenue/km, service quality score, and peak-hour performance, transforming raw booking data into actionable operational insights.
- Designed dynamic dashboards with **DAX-**driven measures**,** date hierarchies, slicers, KPI cards, and interactive visualizations, enabling users to drill into revenue, booking frequency, payment methods, vehicle categories, and cancellation patterns.

AI Video & Meeting Intelligence Assistant:

- Engineered an end-to-end **AI Video/Meeting Assistant** that processes YouTube/local recordings through audio extraction, format conversion, chunking, and automated transcription.
- Integrated **Whisper and Sarvam AI** for English/Hinglish transcription and leveraged **Mistral via LangChain** to generate summaries, titles, action items, key decisions, and open questions.
- Implemented a **RAG-**based conversational layer using ChromaDB and embeddings, allowing users to query the processed meeting/video content through an interactive Streamlit application.

AI-Powered Song Information & Lyrics Agent:

- Built a tool-augmented **LangChain + Mistral AI agent** that accepts a song query and orchestrates external tools to retrieve structured information such as artist, album/movie, music director, lyricist, release year, duration, and lyrics.
- Integrated **Tavily** for web-based retrieval of artist, album/movie, music director, and duration information, while leveraging **LRCLIB** for song metadata and synchronized lyrics, combining specialized sources into a unified response.
- Designed the agent with explicit tool-use instructions and safeguards against hallucination, returning information in a **consistent structured format** and marking unavailable details rather than inventing them.

Portable Real-Time Aerosol Characterization Using Digital Inline Holography & Edge AI :

- Developing a portable digital inline holography (**DIH**) system for real-time, in-situ aerosol characterization, targeting particle size, morphology and spatial information.
- Integrated a coherent laser, camera, flow channel, blower and environmental sensors with an **NVIDIA Jetson Orin Nano** for embedded data acquisition, and built a **PySide6 GUI** for hologram acquisition, visualization, recording and sensor-data logging.
- Working toward numerical hologram reconstruction and **YOLO-based particle morphology** classification for edge deployment; designed a compact, battery-powered, 3D-printed enclosure for field portability.

SKILLS

- **Programming & Data Science:** Python, NumPy, Pandas, Matplotlib, Scikit-learn, Data Preprocessing, Machine Learning, Deep Learning, PyTorch, Optimization.
- **Software & Tools:** FastAPI, PostgreSQL, Power BI, DAX, GitHub, Jupyter, VS Code, Streamlit, Pydantic, LangChain, Generative AI (GenAI), Langgraph, Agentic AI.
- **Imaging & Embedded Systems:** Digital Inline Holography, OpenCV, PySide6, NVIDIA Jetson Orin Nano, Linux/Ubuntu, GPIO, I²C.

POSITIONS OF RESPONSIBILITY

- **Class Representative** -- B.Tech batch (BKIT) and M.Tech batch (IIT Hyderabad); sole point of contact between the student batch and faculty/administration across both programs.
- **Placement Coordinator**, IIT Hyderabad -- coordinated placement-related activities and communication between students and the placement office.
- **Volunteer, International Relations Cell**, IIT Hyderabad -- supported international relations cell activities and initiatives.
- **Fest Event Coordinator**, B.Tech -- coordinated event logistics and execution for the college fest.

HOBBIES & INTERESTS

- **Sports:** Playing and watching **Cricket & Kabaddi**
- **Mobile Gaming** :Free Fire